

# **MRA Project**

## **Part-A, Milestone 1**

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# Agenda

- ▶ **Problem Statement**
- ▶ **Executive Summary**
- ▶ **Exploratory Analysis**
- ▶ **Customer Segmentation using RFM**

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# Problem Statement

An automobile parts manufacturing company has collected data of transactions for 3 years. They do not have any in-house data science team; thus they have hired you as their consultant. Your job is to use your magical data science skills to provide them with suitable insights about their data and their customers.

- **Agenda & Executive Summary of the data**

- Contents of the presentation and Problem statement
- About Data (Info, Shape, Summary Stats, your assumptions about data)

- **Exploratory Analysis and Inferences**

- Univariate, Bivariate, and multivariate analysis using data visualization
- Weekly, Monthly, Quarterly, Yearly Trends in Sales
- Sales Across different Categories of different features in the given data
- Summarize the inferences from the above analysis

- **Customer Segmentation using RFM analysis (make 4 segments)**

- Tool used, all parameters used and assumptions made
- Output table head
- If KNIME used, Workflow image to be put

- **Inferences from RFM Analysis and identified segments**

- Who are best customers (at least 5)
- Which customers are at the verge of churning (at least 5)
- Who are lost customers? (give at least 5)
- Who are loyal customers? (give at least 5)

# Executive Summary of Data

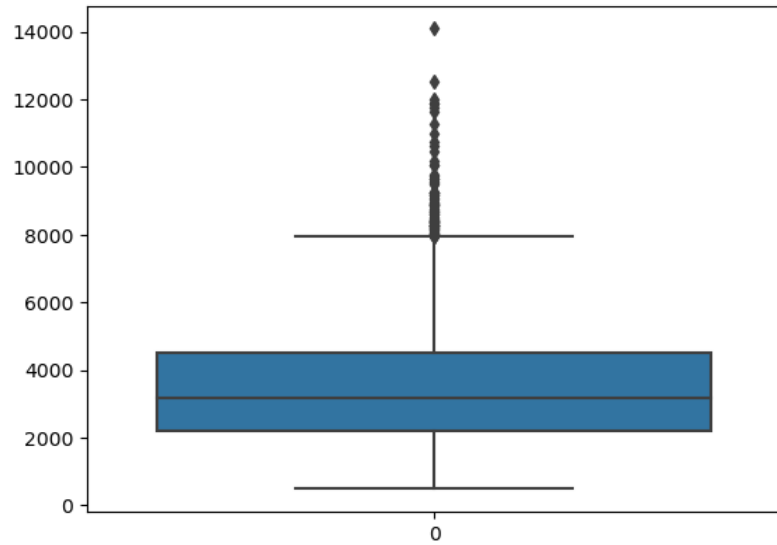
- ▶ The dataset consists of 2747 rows and 20 columns.
- ▶ The data is of 3 years (2018,2019,2020). We will perform Exploratory Analysis first and then we will perform RFM analysis.
- ▶ The data was checked for missing values using Python and there are no missing values in the data. Duplicates, if any were also removed.
- ▶ The following are the features:
- ▶ ORDERNUMBER : Order Number CUSTOMERNAME : customer
- ▶ QUANTITYORDERED : Quantity ordered PHONE : Phone of the customer
- ▶ PRICEEACH : Price of Each item ADDRESSLINE1 : Address of customer
- ▶ ORDERLINENUMBER : order line CITY : City of customer
- ▶ SALES : Sales amount POSTALCODE : Postal Code of customer
- ▶ ORDERDATE : Order Date COUNTRY : Country customer
- ▶ DAYS\_SINCE\_LASTORDER : Days\_ Since Lastorder CONTACTLASTNAME : Contact person customer
- ▶ STATUS : Status of order like Shipped or not CONTACTFIRSTNAME : Contact person customer
- ▶ PRODUCTLINE : Product line – CATEGORY DEALSIZE : Size of the deal based on Quantity and Item Price
- ▶ MSRP : Manufacturer's Suggested Retail Price
- ▶ PRODUCTCODE : Code of Product

# Exploratory Analysis

# Check for Outliers

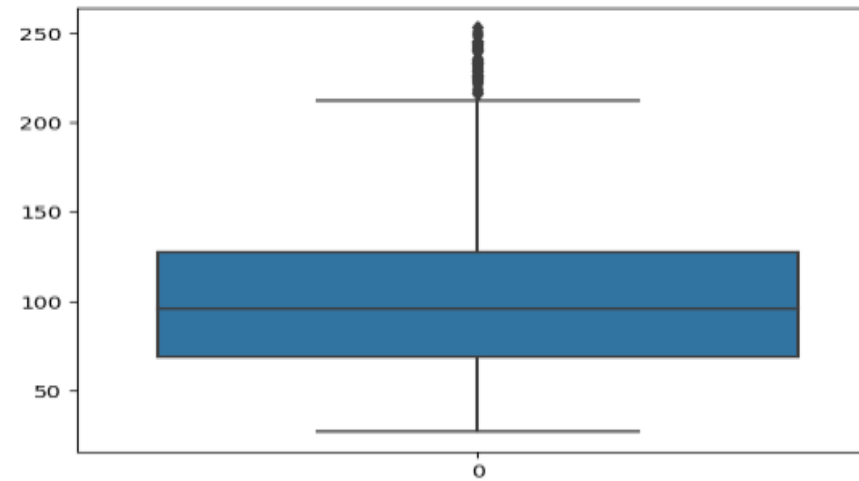
```
In [10]: # Univariate Analysis - Boxplots and Histplots  
sns.boxplot(data=df['SALES'])
```

Out[10]: <Axes: >



```
In [11]: # Univariate Analysis - Boxplots and Histplots  
sns.boxplot(data=df['PRICEEACH'])
```

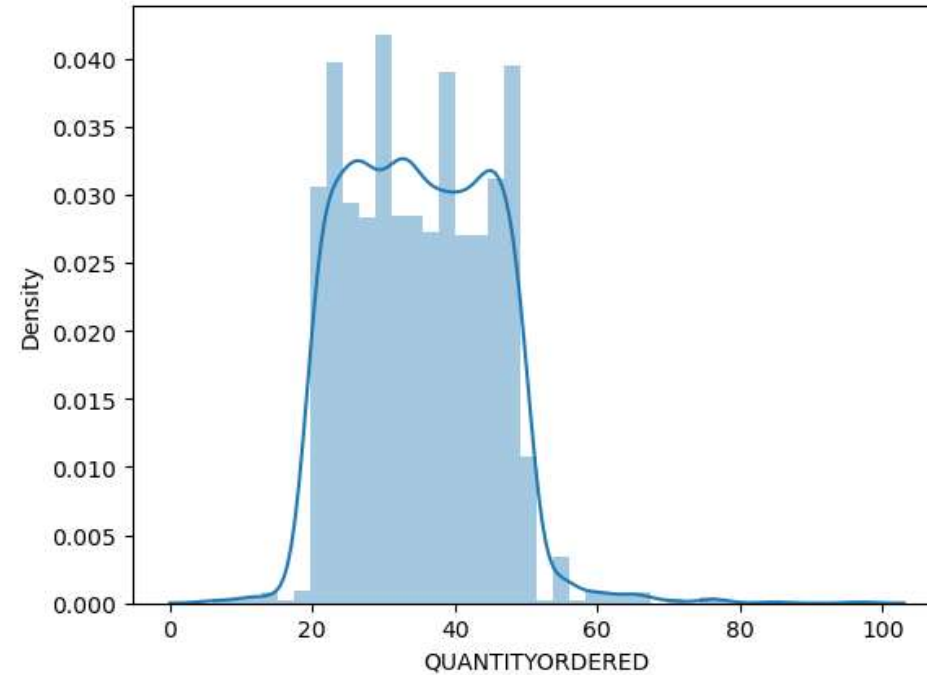
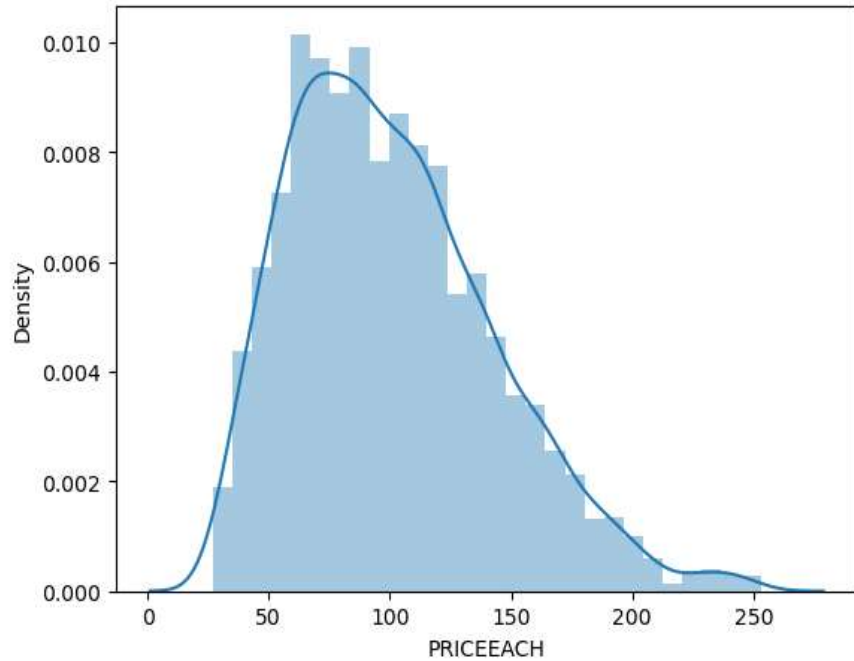
Out[11]: <Axes: >



## Observations:

- ▶ Sales has lot of data points outside the whisker from upper quartile
- ▶ Quantity Ordered , Price Each and MSRP also have some points outside the whisker from upper quartile.

# Distribution Plots - A

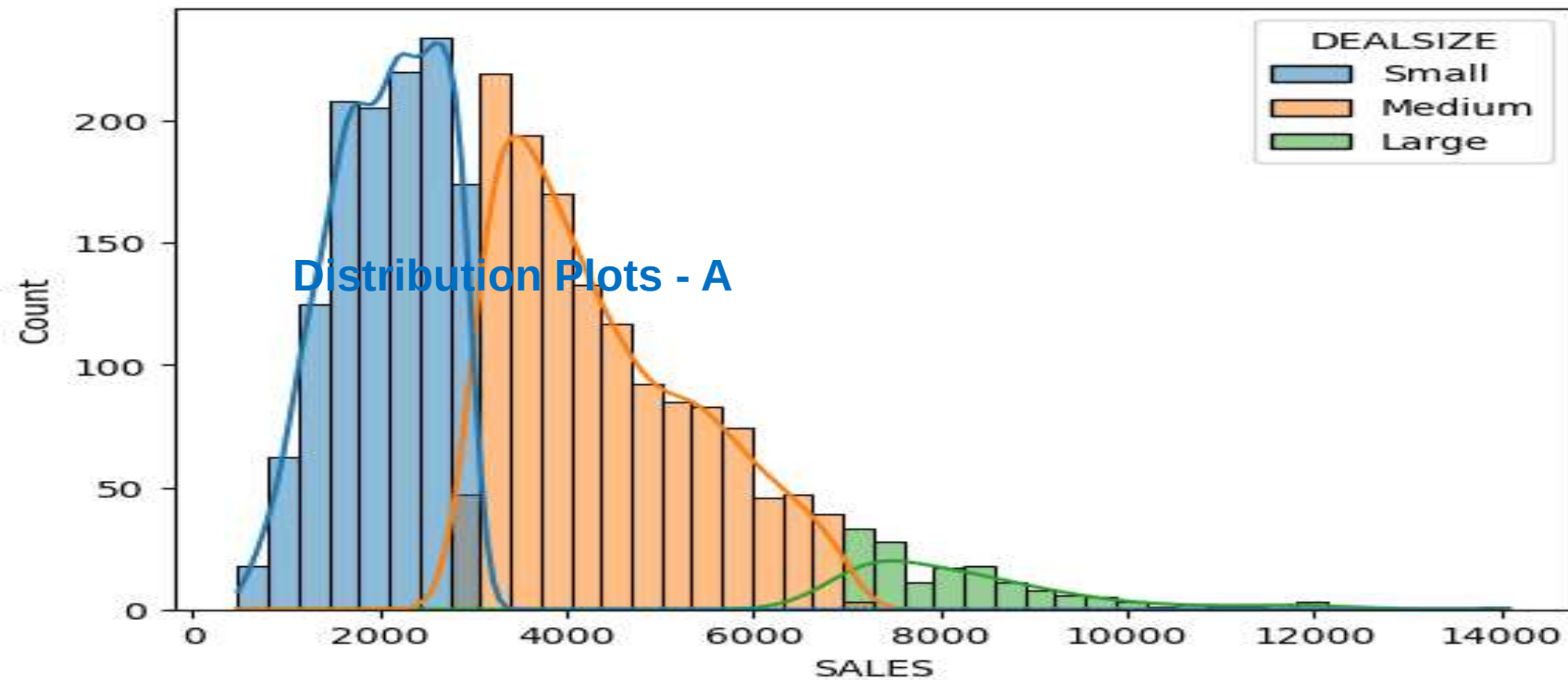


## Observations:

- Price variable has a tail on the right side. Price ranges from approx. 25 to approx. 252.
- Majority of the Customers have Quantity Ordered between 20 and 40.



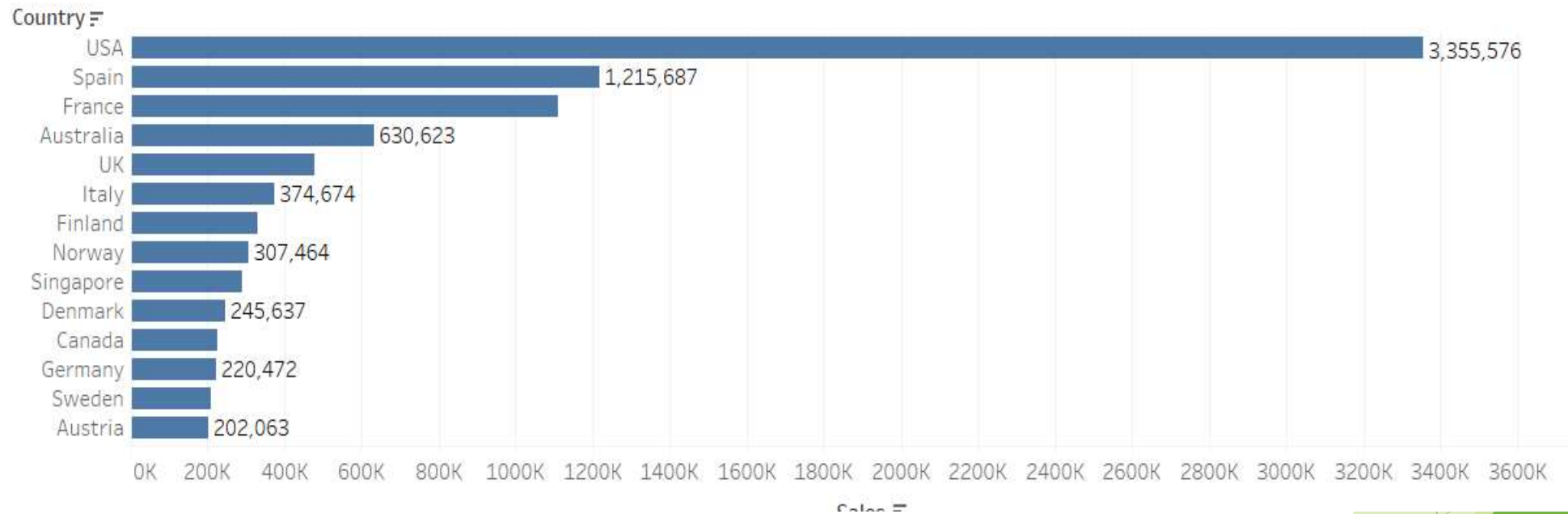
# Distribution Plots - B



- Observation: Majority of Sales come from Small and Medium size deals.

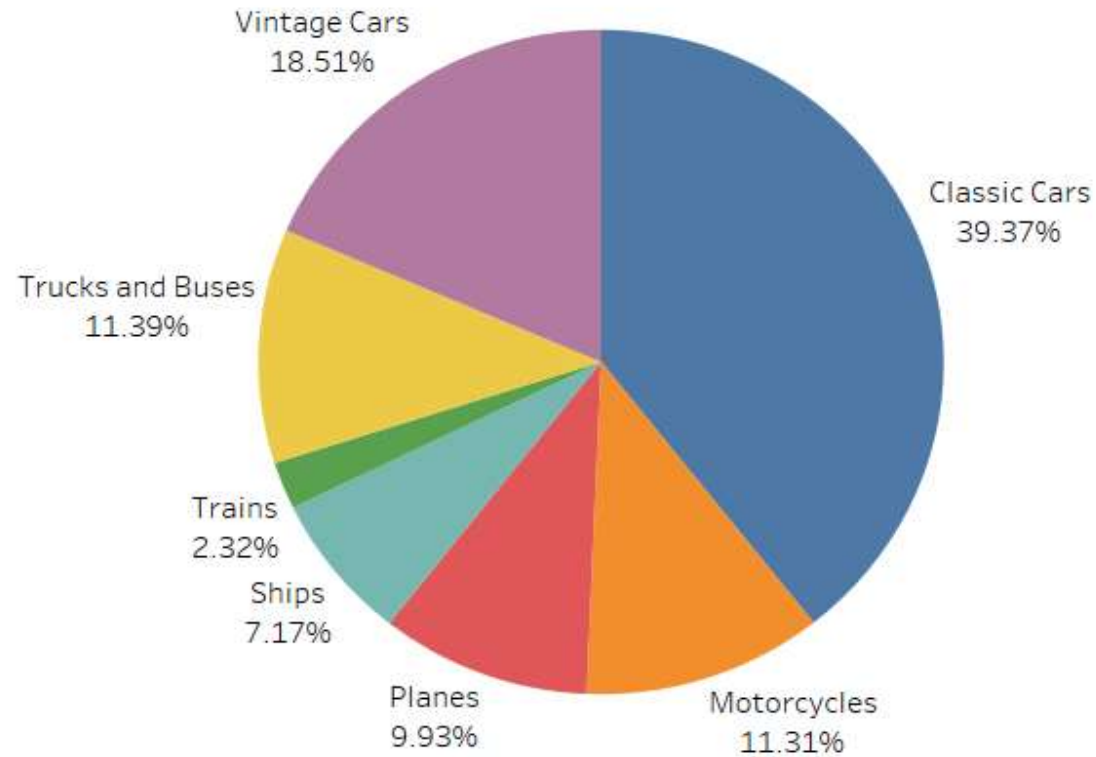
# Sales Across Countries

Bar\_Chart\_Sales\_Across\_Countries



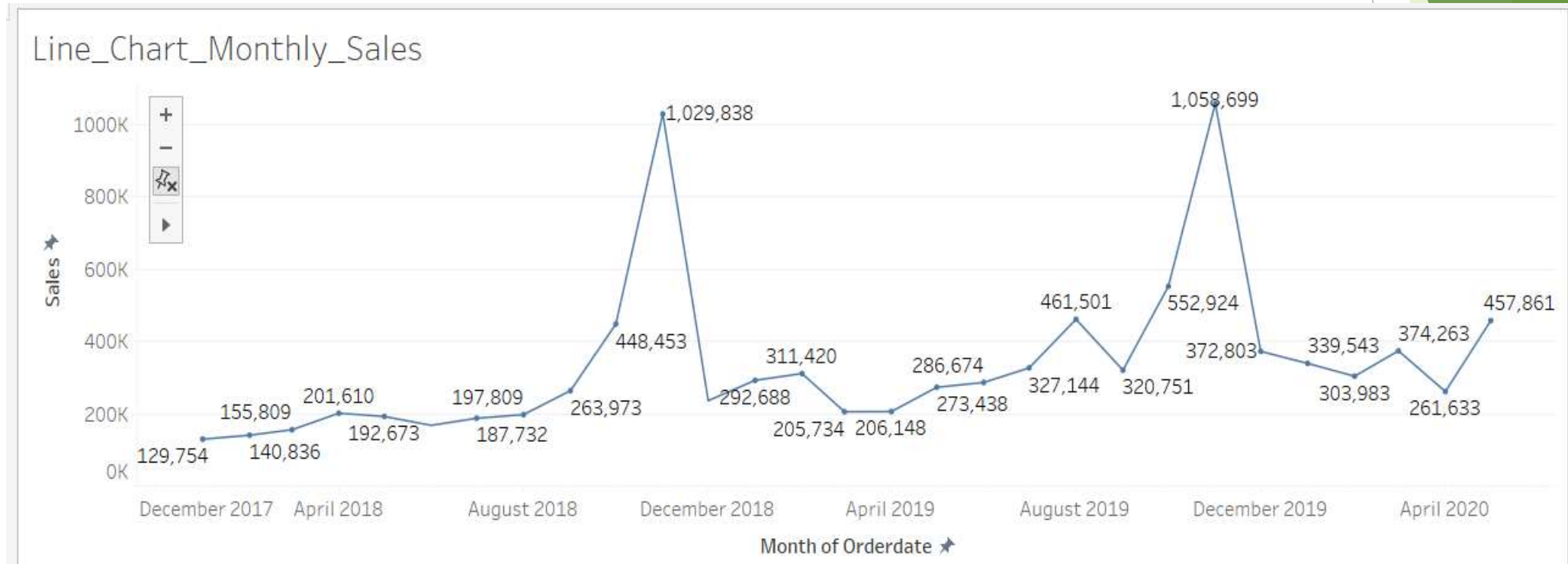
Observations: USA accounts for highest sales followed by Spain, France, Australia and UK.

# Product wise Sales



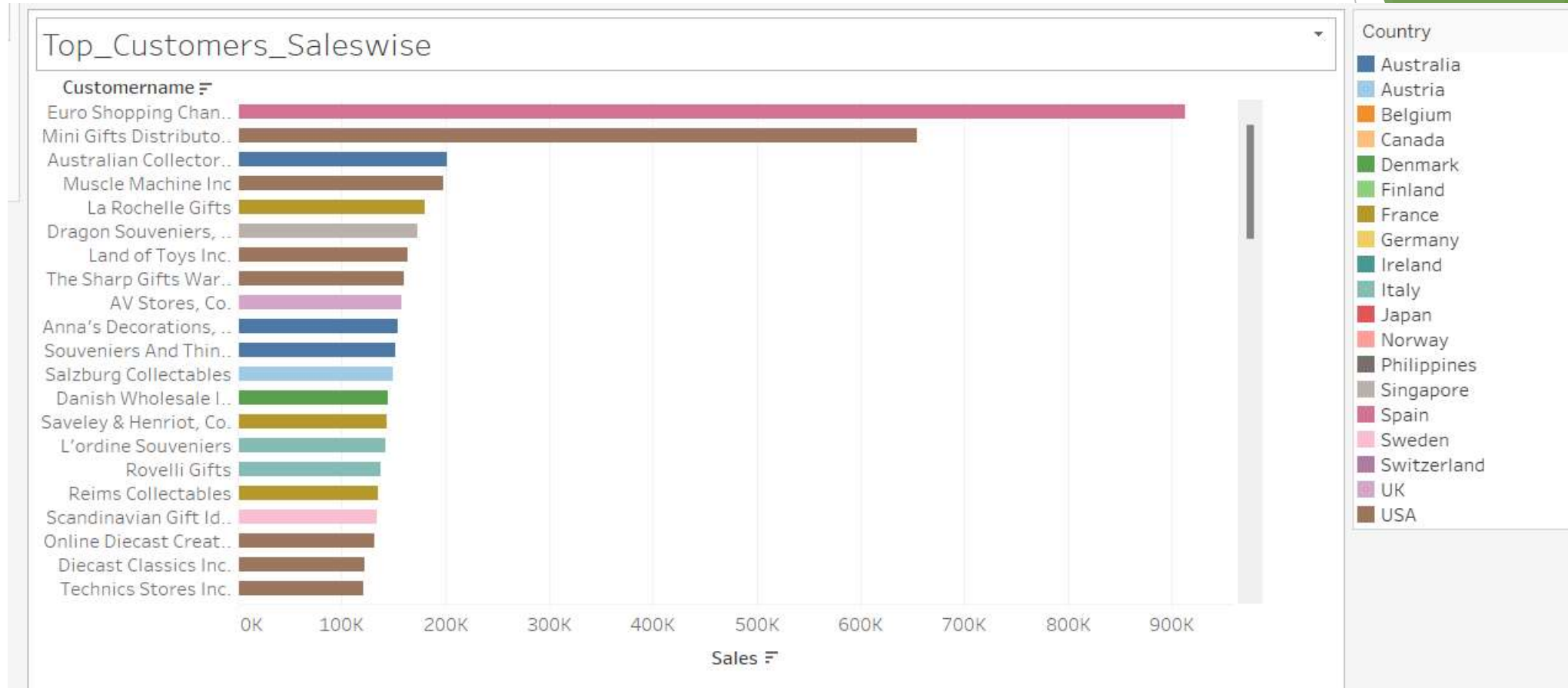
Observation: Classic Cars have maximum sales, followed by Vintage Cars, Trucks & Buses, Motorcycles, planes, Ships and Trains.

# Sales Across Months



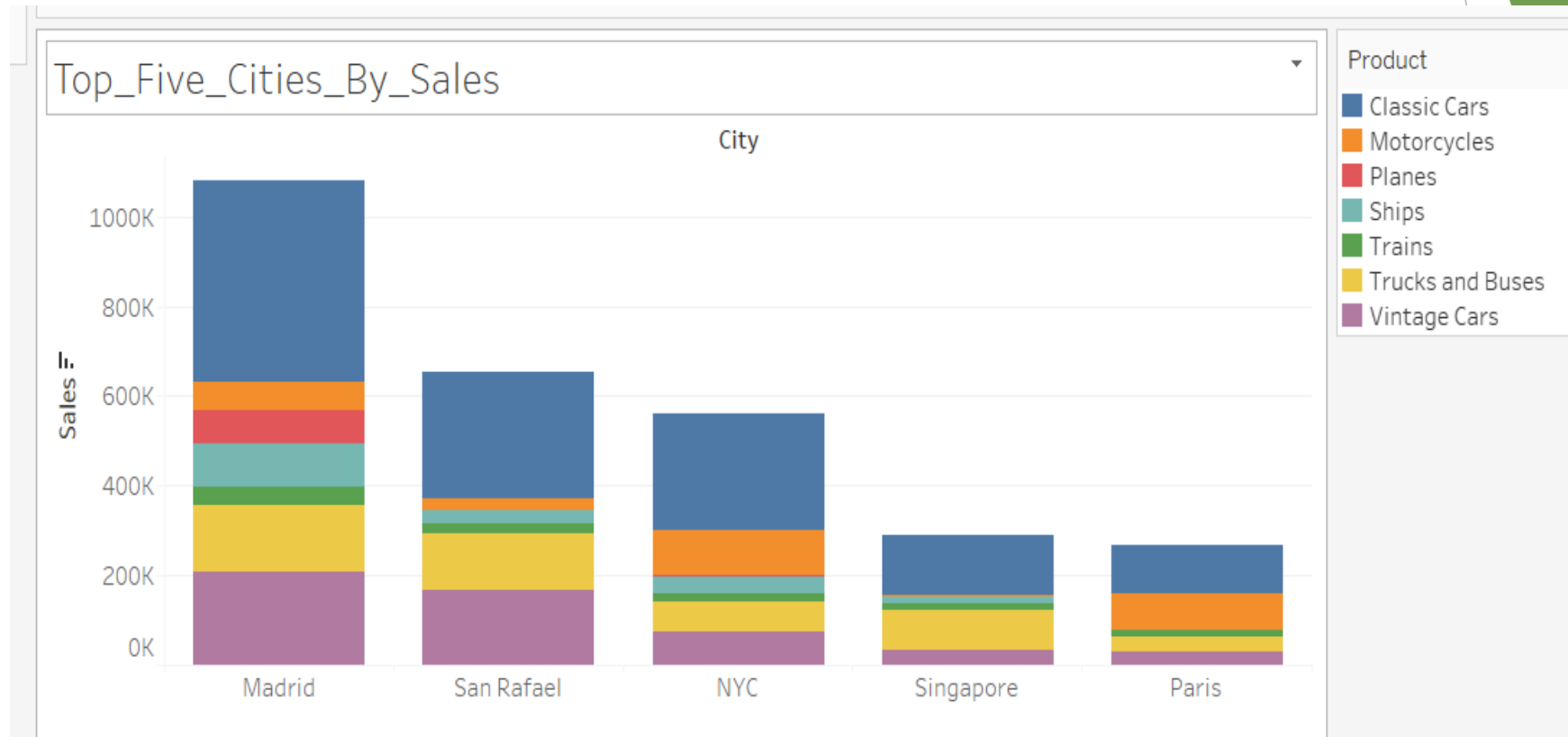
Observation: There is seasonality in the data with high sales in the months of October to December with the peak Sales in the month of November.

# Top Customers Sales wise



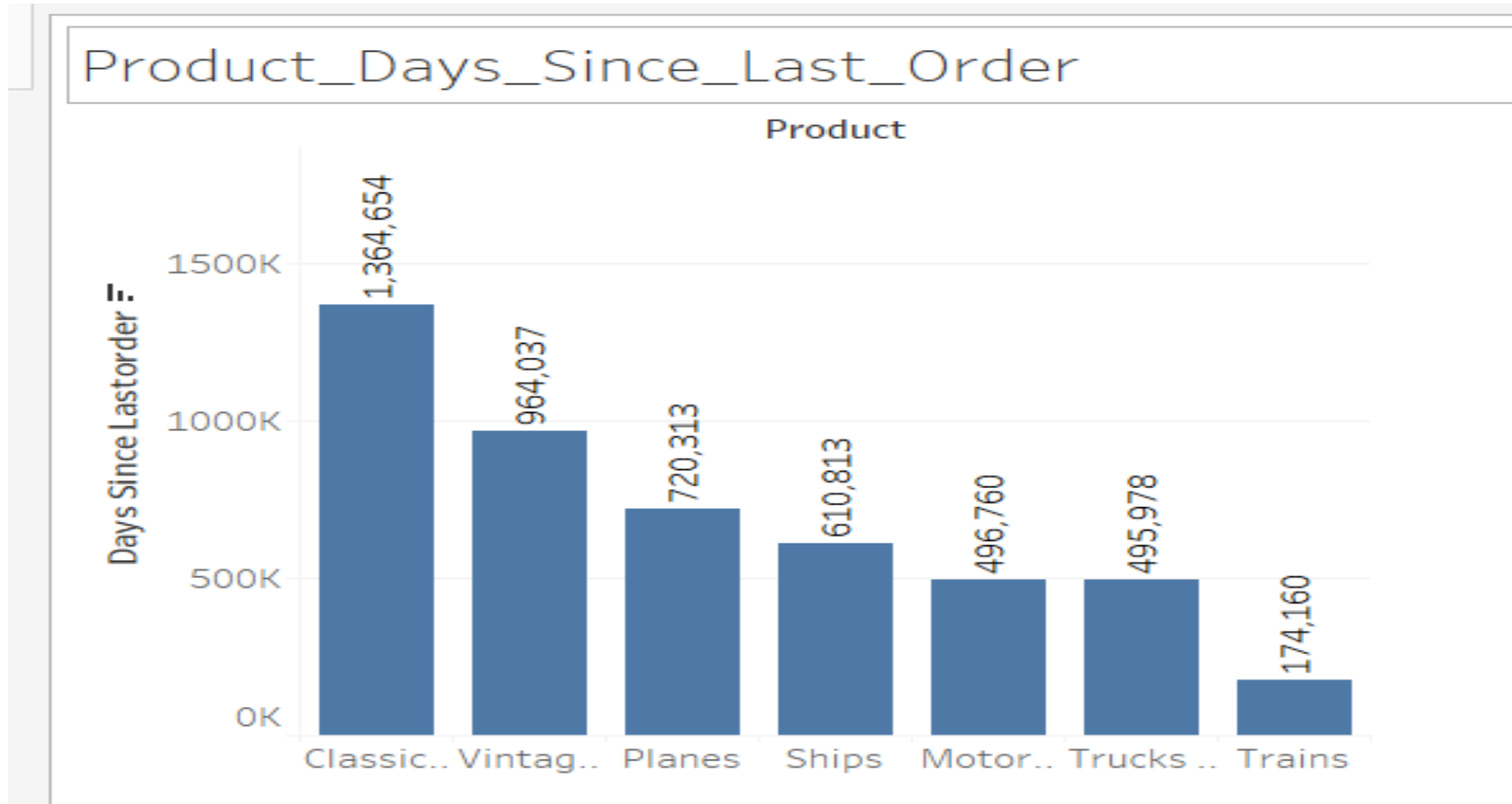
Observation: Top three Customers sales wise are Euro Shopping (Belgium) , Mini Gifts (USA) and Australian Collector (Australia) respectively.

# Top Five Cities by Sales



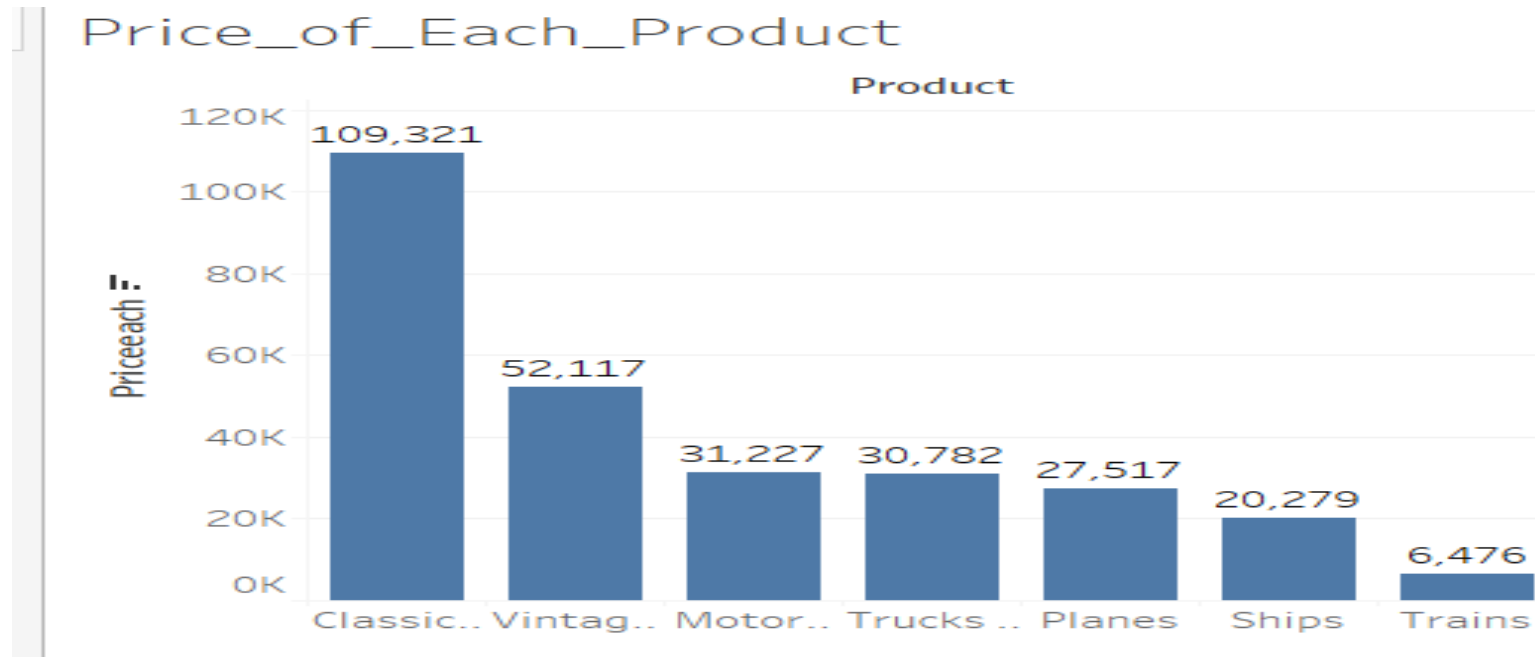
Observation: Top five cities by sales are Madrid, San Rafael, NYC, Singapore and Paris respectively

# Days since last order – Product wise



Observation: Trains is the latest product ordered, followed by Trucks, Motor, Ships, Planes, Vintage Cars and Classic Cars

# Price of Each Product



Observation: Classic Car is the costliest product, followed by Vintage Car, Motor, Trucks, Plain, Ships and Trains



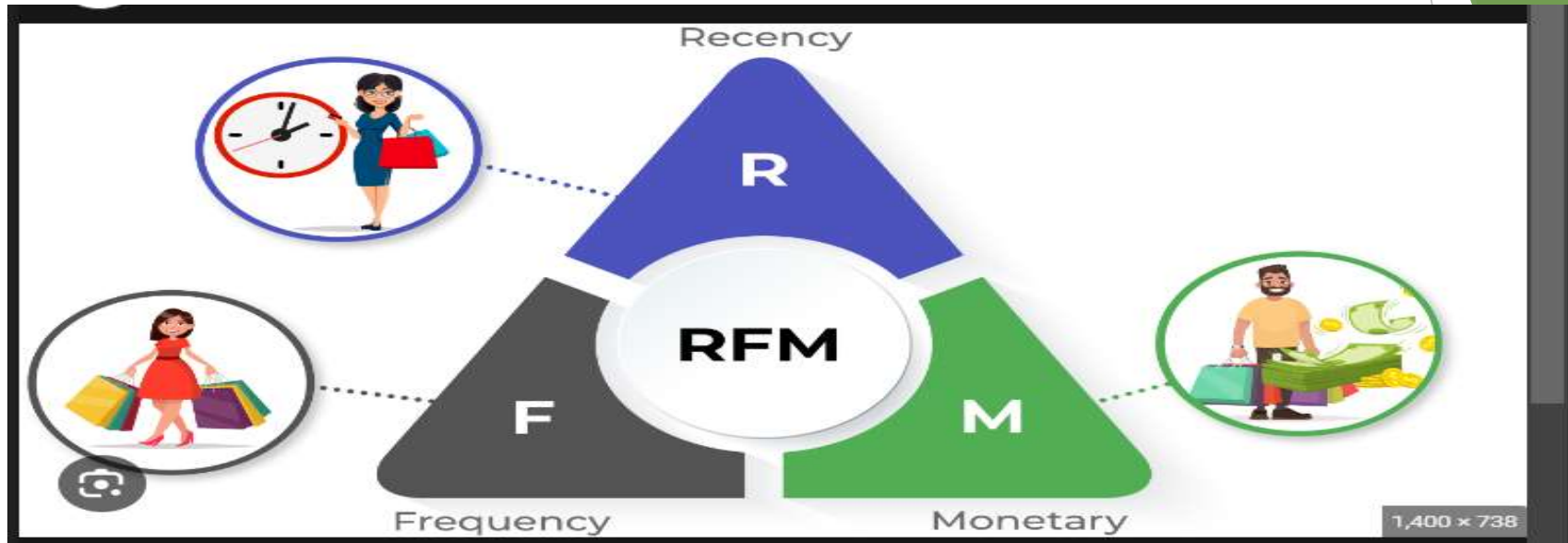
# Key Observations from Data (1)

- ▶ Majority of Sales come from Small and Medium size deals. Very small minority of sales come from Large size deals.
- ▶ USA accounts for the largest volume of sales i.e. 34.38%, followed by Spain, France, Australia and UK.
- ▶ Classic Cars have maximum sales, followed by Vintage Cars, Trucks & Buses, Motorcycles, planes, Ships and Trains.
- ▶ There is seasonality in the data with high sales in the months of October to December with the peak Sales in the month of November.
- ▶ Top three Customers sales wise are Euro Shopping (Belgium) , Mini Gifts (USA) and Australian Collector (Australia) respectively.

## Key Observations from Data (2)

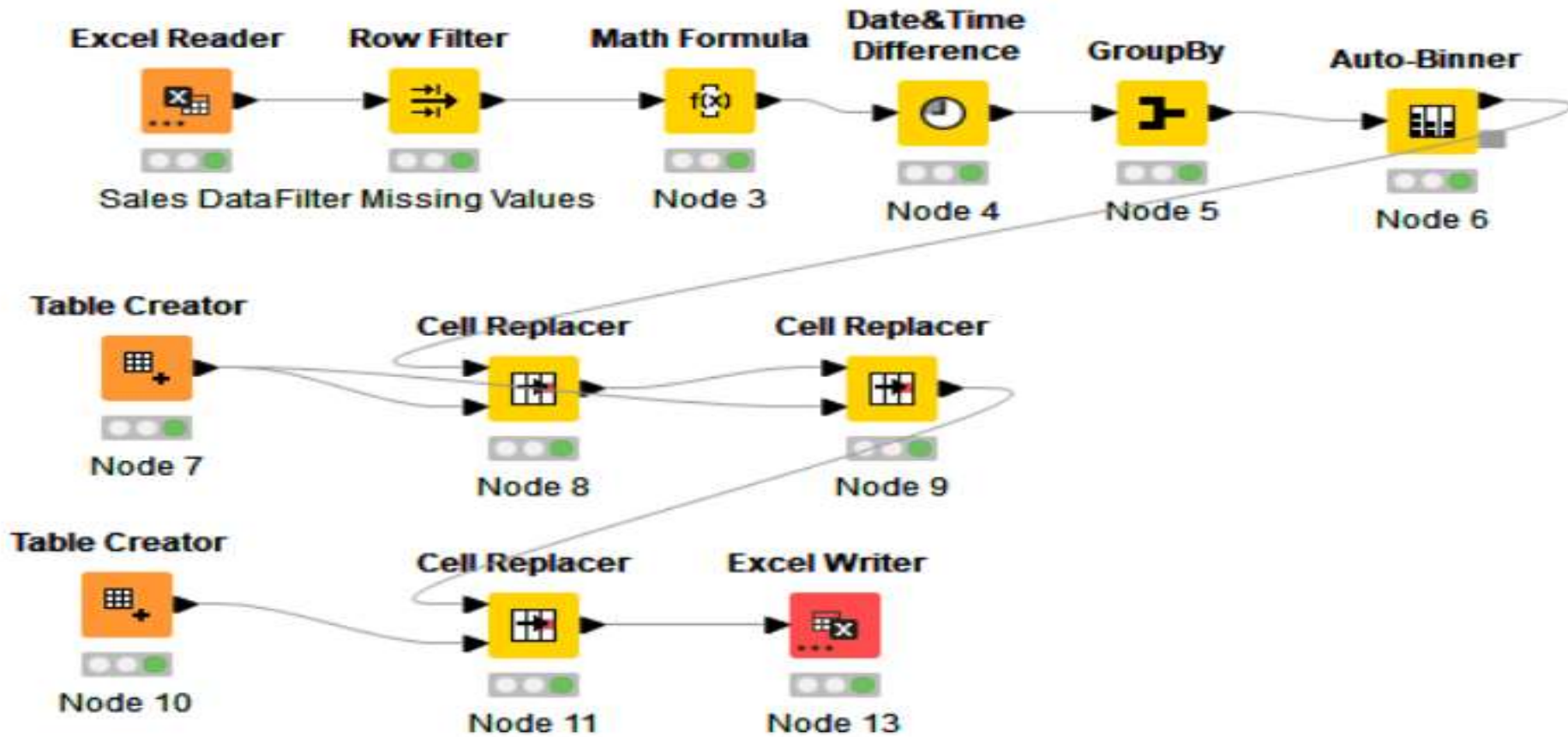
- ▶ Trains is the latest product ordered, followed by Trucks, Motor, Ships, Planes, Vintage Cars and Classic Cars.
- ▶ Classic Car is the costliest product, followed by Vintage Car, Motor, Trucks, Plain, Ships and Trains.
- ▶ Top five cities by sales are Madrid, San Rafael, NYC, Singapore and Paris respectively.

# Customer Segmentation using RFM



RFM segmentation is a technique to identify sets of customers for distinctive treatment. RFM segmentation allows marketers to target specific groups of customers with promotions that are more significant for their actual behavior – and hence provides much higher proportions of response and increased loyalty.

# KNIME WORKFLOW FOR RFM



# RFM Analysis using KNIME

- ▶ **Customers have been Segmented into 4 segments**
- ▶ **Recency** – 4 (most recent)
- ▶ **Frequency** – 4 (most) Frequent and
- ▶ **Monetary** – 4 (Highest order value)
- ▶ *We have grouped the data by Customer Name.*
- ▶ **Aggregation taken –**
- ▶ Unique Count (Orders) for Frequency
- ▶ Sum of Sales for Monetary
- ▶ Minimum of Days Since Last Order (Max Order date – Order Date) i.e. (30-05-2020 – Order Date)

# RFM Results

| Customer Name                | RFM Score |
|------------------------------|-----------|
| Danish Wholesale Imports     | 444       |
| Diecast Classics Inc.        | 444       |
| Euro Shopping Channel        | 444       |
| La Rochelle Gifts            | 444       |
| Mini Gifts Distributors Ltd. | 444       |
| Reims Collectables           | 444       |
| Salzburg Collectables        | 444       |
| Souvenirs And Things Co.     | 444       |
| The Sharp Gifts Warehouse    | 444       |

## Best Customers? (By RFM Class = 444) (highest recency, highest frequency & highest monetary)

| Customer                    | RFM Score | M(Sum Sales) | F (Orders) | R (Recency in days) |
|-----------------------------|-----------|--------------|------------|---------------------|
| Euro Shopping Channel       | 444       | 862398       | 26         | 0                   |
| Mini Gifts Distributors Ltd | 444       | 591549       | 17         | 1                   |
| La Rochelle Gifts           | 444       | 170085       | 4          | 0                   |
| The Sharp Gifts Warehouse   | 444       | 128106       | 4          | 38                  |
| Souvenirs and Things Co     | 444       | 151571       | 4          | 1                   |
| Salzburg Collectables       | 444       | 132440       | 4          | 13                  |
| Danish Wholesale Imports    | 444       | 117508       | 5          | 45                  |
| Reims Collectables          | 444       | 126759       | 5          | 61                  |
| Diecast Classics Inc.       | 444       | 122138       | 4          | 0                   |

## Customers on the verge of churning (Customers with low recency value)

| Customer                       | RFM Score | M(Sum Sales) | F (days) | Recency (days) |
|--------------------------------|-----------|--------------|----------|----------------|
| ▶ Land of Toys Inc             | 244       | 164069       | 4        | 197            |
| ▶ Saveley & Henriot Co.        | 124       | 134271       | 3        | 455            |
| ▶ AV Stores Co.                | 224       | 157808       | 3        | 195            |
| ▶ Rovelli Gifts                | 224       | 111081       | 3        | 200            |
| ▶ Online Diecast Creations Co. | 224       | 119149       | 3        | 208            |



# Lost Customers

(Customers with low recency, frequency & monetary values)

| Customer                       | RFM Score | M(Sum Sales) | F (days) | Recency (days) |
|--------------------------------|-----------|--------------|----------|----------------|
| ▶ West Coast Collectables Co.  | 111       | 46085        | 2        | 487            |
| ▶                              |           |              |          |                |
| ▶ Signal Collectables Ltd.     | 111       | 41770        | 2        | 475            |
|                                |           |              |          |                |
| Online Mini Collectables       | 111       | 47567        | 2        | 263            |
|                                |           |              |          |                |
| Iberia Gift Imports            | 111       | 54724        | 2        | 237            |
|                                |           |              |          |                |
| Double Decker Gift Stores Ltd. | 111       | 36019        | 2        | 494            |
|                                |           |              |          |                |

## Loyal Customers (Customers with high frequency value)

| Customer                       | RFM Score | M(Sum Sales) | F (days) | Recency (days) |
|--------------------------------|-----------|--------------|----------|----------------|
| ▶ Mini Gifts Distributors Ltd. | 444       | 591549       | 17       | 1              |
| ▶ Euro Shopping Channel        | 444       | 862398       | 26       | 0              |
| Reims Collectables             | 444       | 126759       | 5        | 61             |
| Danish Wholesale Imports       | 444       | 117507       | 5        | 45             |
| Australian Collectors Co.      | 344       | 181957       | 5        | 183            |

**Thank You**